

WearUS: A Unified Framework for Wearable A-Mode Ultrasound Pretraining and Downstream Transfer

Tenth Weekly Report

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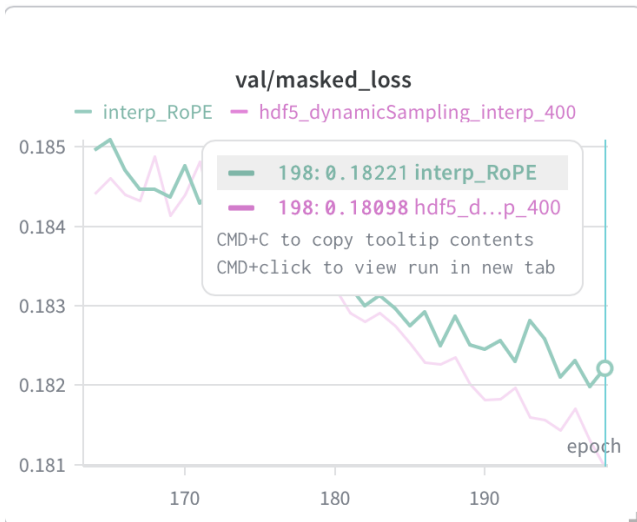
• NORM Mode Implementation:

- In NORM mode, the loss is calculated between the model's reconstruction and the **target normalized per patch** (standardized to zero mean and unit variance within each patch).
- This follows the original MAE approach for image representation learning, adapted here for A-mode signals.

• Intuition and Rationale:

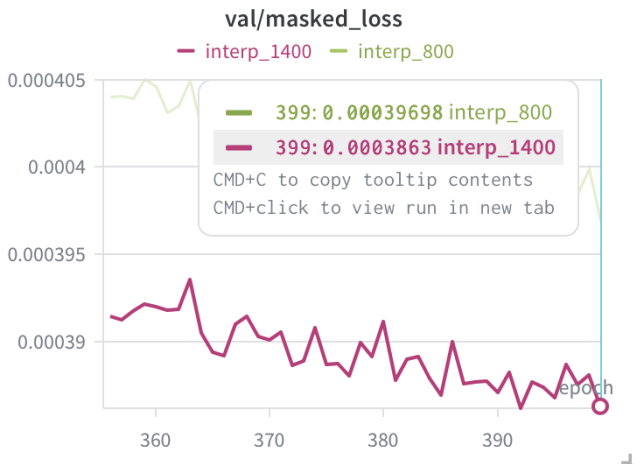
- Useful for understanding **high-level features**: by normalizing, we focus on the local structure rather than the absolute intensity.
- Prevents "model laziness": in standard mode, the model might converge to predicting zero for low-amplitude segments (near-zero features) to minimize loss easily.
- Forces the model to reconstruct the **shape and frequency** of the signal even when the real values are small.

CT-RoPE vs. RoPE



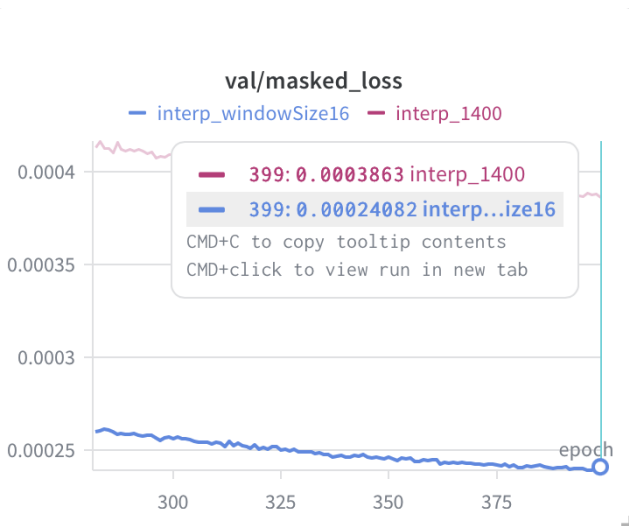
Very marginal improvement in loss

interp1400 vs interp800



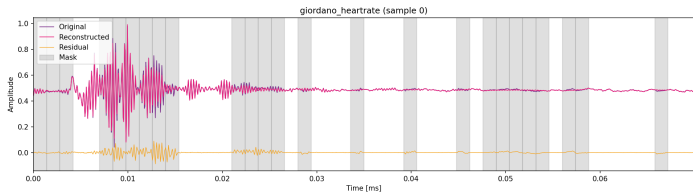
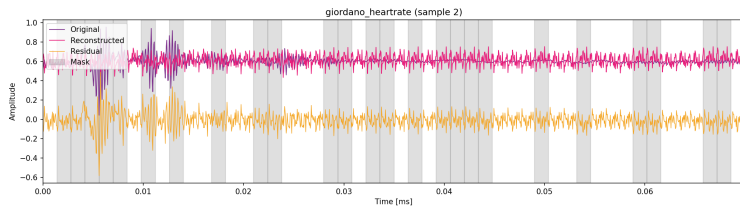
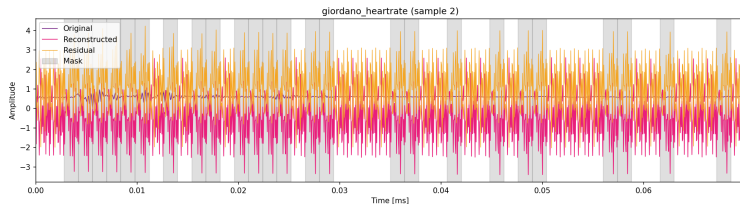
No improvements

S100 vs S50



Big improvement

training history



Implementation Details

- In the encoder, only visible patches enter.
- Then in the decoder, the masked tokens are reinserted in their positions and reconstructed.
- I changed the masking from zero masking to a **learned mask token** in order to be aligned with the original MAE paper.

Masked Autoencoders Are Scalable Vision Learners

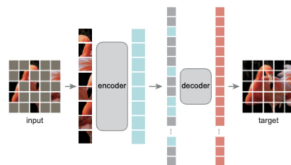
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Facebook AI Research (FAIR)

Abstract

This paper shows that masked autoencoders (MAE) are scalable self-supervised learners for computer vision. Our MAE approach is simple: we mask random patches of the input image and reconstruct the missing pixels. It is based on two core designs. First, we develop an asymmetric encoder-decoder architecture, with an encoder that operates only on the visible subset of patches (without mask tokens), along with a lightweight decoder that reconstructs the original image from the latent representation and mask



✓ 19 Dec 2021